INDUSTRIAL SPOTLIGHT · MULTI-SENSOR SYSTEM

Multi-sensor Registration for an Autonomous Forklift

Apply coordinate registration across sensors and connect perception to safety decisions and the vehicle system.

EVIDENCE STATE **Ongoing**	PERIOD **2024 - present**
PORTFOLIO TIER **Industrial Spotlight**	PRIMARY CAPABILITY **Sensor Fusion & Localization**

Connected ToF-RGB-SAM3 registration, SICK TiM LiDAR and NAV350 3D PCD processing, robot localization, sensor fusion, safety policy, and Zenoh publication into one integration flow.

Problem and system boundary

01 / SENSOR COORDINATE CHAIN

Sensor coordinate chain

Connected RGB, ToF, LiDAR, and NAV350 to vehicle coordinates and robot pose.

PROBLEM

Sensor streams arrived in different coordinates and cycles and needed a common flow usable by safety decisions.

Connected ToF-RGB-SAM3 registration, SICK TiM LiDAR and NAV350 3D PCD processing, robot localization, sensor fusion, safety policy, and Zenoh publication into one integration flow.

PUBLIC BOUNDARY

The public clip and photos are test-site runs; they do not claim production or customer outcomes.

Vehicle control, sensing, safety, and field-validation owners integrate the system jointly.

EVIDENCE STATE

Ongoing / Screen recording of the DOTORI vision-servoing test (3D view, camera feed, signal plots).

Owned decisions and implementation

MY ROLE

Owned ToF-RGB-SAM3 registration; SICK TiM LiDAR and NAV350 3D PCD processing; robot localization; sensor fusion and safety-policy decisions; and publishing results through Zenoh.

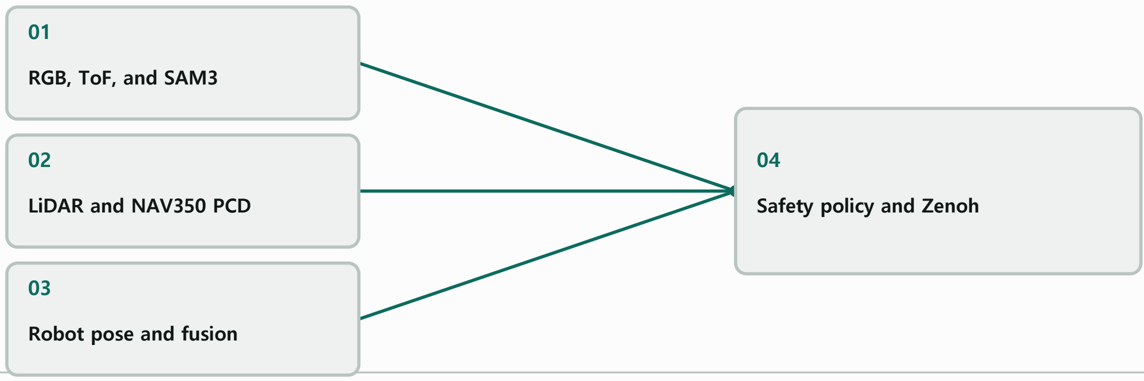
02 / PERCEPTION TO POLICY

Perception to policy

Connected registration and pose outputs to safety-policy decisions and Zenoh messages.

SYSTEM RELATIONSHIP

Multi-sensor convergence to safety policy



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

C++23 / ROS 2 / Zenoh / ToF / RGB / SAM3 / SICK TiM LiDAR / NAV350

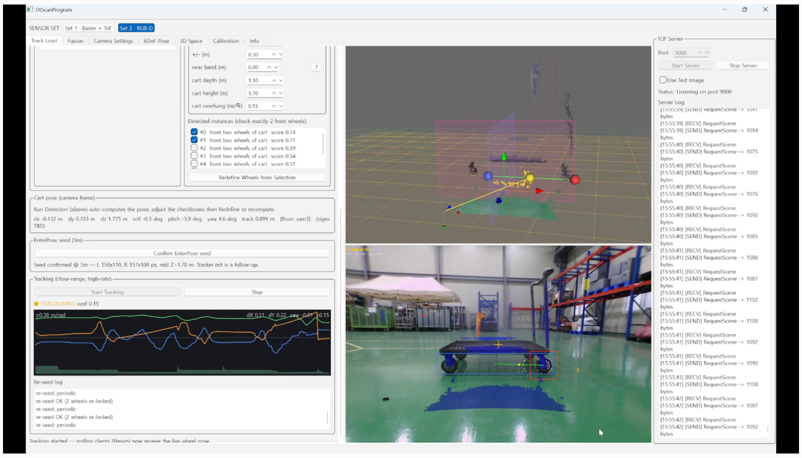
Verification and evidence ledger

03 / INTEGRATION EVIDENCE

Integration evidence

Compared sensor alignment, point clouds, pose, and policy output in integration and field tests.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

VIDEO / PUBLICLY INSPECTABLE

unmanned-forklift-clip-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

unmanned-forklift-poster-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

unmanned-forklift-gallery-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

unmanned-forklift-gallery-02

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

unmanned-forklift-gallery-03

Approved derivative; caption in portfolio data.

Role, team result, and limits

04 / FIELD BOUNDARY

Field boundary

Field validation is not production operation, deployment success, or a customer outcome.

MY ROLE

Owned registration, PCD, localization, sensor fusion and safety policy, and Zenoh output.

Owned ToF-RGB-SAM3 registration; SICK TiM LiDAR and NAV350 3D PCD processing; robot localization; sensor fusion and safety-policy decisions; and publishing results through Zenoh.

TEAM RESULT

The team performed system integration and field validation. This is not presented as production operation, deployment success, or customer

Registration outputs, 3D point clouds, robot pose, safety decisions, and Zenoh messages were checked through integration and field validation.

LIMITATIONS

The public clip and photos are test-site runs; they do not claim production or customer outcomes.

Screen recording of the DOTORI vision-servoing test (3D view, camera feed, signal plots).

COLLABORATION BOUNDARY

Vehicle control, sensing, safety, and field-validation owners integrate the system jointly.

Connected ToF-RGB-SAM3 registration, SICK TiM LiDAR and NAV350 3D PCD processing, robot localization, sensor fusion, safety policy, and Zenoh publication into one integration flow.

uiop3847@naver.com

Recap and joint-development invitation

Apply coordinate registration across sensors and connect perception to safety decisions and the vehicle system.

PRIMARY CAPABILITY

Sensor Fusion & Localization / 3D Geometry & Registration

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Vehicle control, sensing, safety, and field-validation owners integrate the system jointly.

uiop3847@naver.com
rafaam11.github.io